Merged Exam Questions

1. These are not exam
2. BACHELOR OF SCIENCE (MATHEMATICS AND COMPUTER SCIENCE)
3. **Instructions:** Answer
4. QUESTION
5. Distinguish
6. Explain
7. c) State four types of keys that can be used in a relation. (4 marks)
8. Here's
9. f) Draw an E-R Diagram for Ruiru Hospital that captures the following entities: Patients.
10. Doctors, Assistants and Wards. (5 marks)
11. g) State three database management systems. (3 marks)
12. The provided input consists
13. a) Jack wants to design tables to be used in a given database. Explain three types of entity
14. relationships that he can use in the design. (6 marks)
15. Models are
16. Analyze and discuss various
17. c) State six advantages of a computerized database over the traditional file system.
18. a) Show, giving appropriate SQL commands, step by _step how you would go about: i. Creating a database called Machakos with a table named Employee (attributes:
19. Code, No, Name and Rate) in SQL. (4 marks)
20. Add two records to
21. iii
22. **Merged question:**
23. design cycle that he is supposed to follow. (6 marks)

24. **

25. Describe the five

26. each other. (10 marks)

27. **Merged

28. c) Erick, a database administrator is supposed to protect the database from various threats.

29. Advise him on three methods he can use to perform this task. (6 marks)

30. b) Discuss three types of relationships that can be implemented in databases using well

31. labelled diagrams. (6 marks)

32. **Merged question:**

33. **Merged question

34. e. Briefly describe three reasons for securing a database (6 marks) f. Draw an E-R Diagram for MUC that captures the following entities: Students. Lecturers,

35. Library and Computer Lab. (5 marks)

36. State

37. **

38. Explain three types

39. Here'

40. **

41. Please

42. **Merged question:**

43. other. (10 marks)

44. b. Distinguish between the logical and physical database schema (4 marks) d. Erick, a database administrator is supposed to protect the database from various threats.

45. discuss the process of normalization from first up to the third normal forms (10 marks) (c) A systems administrator wants to organize files in the machines that exist in their work place.

46. Explain five file organization

47. i) Data integrity

48. Describe

49. f) Draw an E-R Diagram for MUCthat captures the following entities: Students. Lecturers,

50. g) State three rolesof adatabase administrator. (3 marks) -- 1 of 2 -- Examination Irregularity is punishable by expulsion Page 2 of 3

51. Show, giving

52. Rate) in MySQL. (4 marks)

53. database development cycle that he is supposed to follow. (10 marks)

54. (10 marks) (c) A systems administrator wants to organize files in the machines that exist in their work

55. Explain the

56. **

57. b) Describe three levels of abstraction in the DBMS? (6 marks)

58. c) Highlight three types and use and application of various Relational Database Keys

59. d) Describe :- i. The best practices in database security that can help keep your databases

60. safe from attackers. (4 marks)

61. Transform the given table

62. a) Define a CLAUSE in terms of SQL (2 marks)

63. Analyse and describe

64. c) Explain two disadvantages of ER Model in DBMS (4 marks)

65. d) Discuss the characteristics of a relational table (8 marks)

66. a) Define a distributed database system (1 mark)

67. b) Using a relational illustration, explain functional dependency in the DBMS (3 marks)

68. c) Describe the Importance of Data Modeling in DBMS (4 marks)

69. d) Use the table below to answer questions that follow. salesman_id name city commission 5001 James Kamau Nairobi 0.15 5002 Nancy Okoth Kisumu 0.13 5005 Alex Lewa Mombasa 0.11 5006 Mercy Mueni Kisumu 0.14 5007 Paul Adam Nakuru 0.13 5003 Derrick Pemba

Kampala 0.12

70. i. Write an SQL statement to create a database “Machakos Merchants” that will be used

71. for the above table. (2 marks)

72. ii. Write an SQL statement that could be used to create the above table. (4 marks)

73. iii. Write a SQL statement to display all the information of all salesmen. (2 marks)

74. iv. Write a SQL statement to display those salesmen who come from Kisumu city.

75. a) Define a RDBMS (2 marks)

76. b) Explain three characteristics of Network Database Management System (DDBMS)

77. **Merged question:**

78. c) Using illustrations, Describe three types of relationships in the DBMS (5 marks)

79. d) Analyse four states of transaction management in DBMS (8 marks) -- 2 of 2 --

80. b) Given user A and user B are transacting and updating record in the same table, with the aid

81. of a diagram, simulate a lost update as a result of an inconsistent read. (6 marks)

82. c) Discuss two Database authentication methods in a Local Area Connection. (4 marks)

83. d) While setting REMOTE_LOGIN_PASSWORDFILE, several aspects must be taken into

84. account. Discuss three which can be set on this parameter and their implication. (6 marks)

85. In a busy network

86. marks)

87. f) You have been hired as a database administrator in a data center. The center is being set up for the first time. Highlight four factors you would consider while creating a database.

88. Discuss

89. Discuss the multifaceted role

90. methods (6 marks)

91. Explain the

92. List

93. b) Propose two methods to solve user authentication flaws in database management systems.

94. c) List four database system actions which can be assigned to a subject to execute on an object.

95. d) List four Mechanisms to facilitate restoration of a database quickly and accurately after loss

96. These are not complete

97. While

98. Explain five backup facilities

99. **Merged question:**

100. c) Database performance is determined by several parameters. Determine four aspects of

101. database system that a database administrator needs to monitor. (4 marks)

102. d) Define the following terms as they relate to database administration (6 marks) i. Tablespace ii. Instance -- 2 of 3 -- Examination Irregularity is punishable by expulsion Page 3 of 3 iii. Data dictionary iv. Redo log v. Shema vi. Timestamp

103. a) Databases are bound to fail due to various factors. Discuss four options that a database administrator can choose to respond to the failure, Indicating the circumstances most suitable for each. (8

104. **Merged

105. c) Highlight four levels of locks. (4 marks)

106. d) With the aid of a diagram, show how roll forward database recovery works. (4 marks) -- 3 of 3 --

107. a) Distinguish between the following terms and concepts. (4 marks) (i) Data Physical Format and Data Logical Format as applies to data dependence. (ii) Data Inconsistency and Data Anomalies as applies to data redundancy.

108. b) “The Database Administrator (DBA) must be prepared to recover data to a usable point, no matter what the cause, and to do so as quickly as possible.”. Explain the three types of

109. data recovery referred to in this statement. (6 marks)

110. c) When asked by their lecturer why they think studying relational algebra and relational calculus is important in Advance Database Systems, the following students responded as

111. follows. (4 marks) Janet Relational Algebra provides the relational model with a flexible way

to query a database. Brian Relational Calculus is the foundation of Query-By-Example. Justify these statements.

112. d) Outline three common characteristics that data models must have in order to be widely

113. accepted. (3 marks)

114. e) “When implementing a physical database from a logical data model, there is need to consider how the database will perform when applications make requests to access and modify data.”. Discuss this statement with emphasis on techniques that are employed to

115. allow data to be accessed in the database more rapidly. (4 marks)

116. f) Explain two ways in which views can be used to implement data security. (2 marks) -- 1 of 3 -- Examination Irregularity is punishable by expulsion Page 2 of 3

117. g) A relational database can be used for text retrieval as follows. The database needs two tables. The first, Documents, has two attributes, DID and Text; one is a unique document identifier, the other is the full text of a document. The other, Words, also has two attributes, Word and DID; the first is a word (that is, a string), the other is the identifier of a document containing that word. (i) Explain how a Boolean query on the text of a document could be formulated in

118. SQL on such a database, giving examples. (4 marks) (ii) Would you expect such querying to be reasonably fast? Why? Consider possible

119. costings and possible queries. (3 marks)

120. a) Outline two advantages and two disadvantages of object-oriented database model.

121. Here's the

122. i) Explain any two important functions it performs that guarantee the integrity and consistency of data in the database. ii) One technical and one non-technical DBMS strategy an organization can adopt in establishing a usable database environment. iii) Two considerations needed in understanding the DBMS requirements and preparation of the environment for the new DBMS in the organization. iv) One benefit and one risk in upgrading the DBMS in organizations database environment.

123. c) Consider the relation R(A,B,C,D,E,F,G,H,I,J,K) which is in First Normal Form (1NF). Suppose its dependencies are A,B!’C B,D!’E,F A,D!’G,H A!’J H!’K.

124. i) Identify the key of R. (2 marks)

125. De

126. a) The following are relations from a database. Use them to answer the questions that

127. follow. (10 marks) Sailors (sid, sname, rating, age) Boats (bid, bname, color) Reserves (sid,

bid, day) Represent the query below in Relational Algebra, Tuple Relational Calculus, Domain Relational Calculus and Structured Query Language (SQL). Find the names of sailors who reserved boat #103.

128. b) Draw an Entity-Relationship (ER) diagram showing entities, attributes, relationships,

129. cardinality and participation. State any assumptions necessary here. (10 marks) A worksite at a particular address has several named workers with tax file numbers and tasks. Building materials of certain types and quantities are delivered by named trucking companies on a date to a supervisor at the worksite; there are several manufacturers of each kind of material, each with their own business name and address. -- 2 of 3 -- Examination Irregularity is punishable by expulsion Page 3 of 3

130. Write SQL statements to

131. i) Updating one or more records (2 marks)

132. ii) Changing attributes characteristics (2 marks)

133. iii) Deleting all rows in from a table (2 marks)

134. iv) Primary key designation (2 marks)

135. b) Explain the significance of the two basic types of Data Control Language statements as

136. used in database security. (4 marks)

137. c) Using an SQL statement, explain two types of privileges that can be granted and revoked

138. These

139. d) Write SQL statements that demonstrate the creation of a trigger and a stored procedure

140. Explain

141. b) Consider database objects A and B and assume that there are two transactions T1 and T2.

142. Produce a

143. object P and Q and then writes objects P and Q. State an example schedule with actions

144. c) Discuss the role of a database administrator enforcing security in database design.

145. d) Explain the difference between the two methods used for accessing relational data from a

146. Java program. (4 marks)

147. BACHELOR OF SCIENCE MATHS AND COMPUTER SCIENCE BACHELOR OF SCIENCE STATISTICS AND PROGRAMMING BACHELOR OF SCIENCE ELECTRICAL ENGINEERING

148. i. Data availability (4 marks)

149. ii. Data validation (4 marks)

150. b) A tennis club uses the following table to record details of players and their coaches. PlayerID Name Ranking CoachID CoachName P001 Kariuki 12 C003 Ben P002 Okoth 3 C013 Mark P003 Tanui 17 C003 Ben P004 Mosioma 9 C006 Titus

151. i. Explain why the above table is not in 3rd normal form. (4 marks)

152. c) The table below records orders for items. Each order is placed on a given date, and may include a variety of items in different quantities. Orders OrderNo ItemNo Description Date Quantity 1 12 Screw 6th jan 100 1 15 bolt 6th jan 50 2 7 Flange Feb 2nd 10 2 15 Bolt Feb 2nd 40 2 12 screw Feb 2nd 80 i. Give an example of an insertion anomaly and an example of modification (update)

153. anomaly that might occur in the above table. (4 marks)

154. ii. Transform the table into 2nd Normal Form (6 marks)

155. a) Explain three stages of data base design (6 marks)

156. c) Distinguish between a distributed database and a distributed database management system.

157. a) Describe three integrity constraints that could be applied to a database management system

158. during its design stage. (6 marks)

159. b) Explain two update anomalies associated with poor database design. (2 marks)

160. Explain four vital

161. expected to perform in his everyday operations (6 marks)

162. d) Describe five advantages of using a database system over the use of spreadsheets. (6 marks) -- 2 of 3 -- Examination Irregularity is punishable by expulsion Page 3 of 3

163. a) Highlight three characteristics of using a primary key during database design (6 marks)

164. a) Describe, using an example, one type of problem that can occur in a multi-user environment

165. when concurrent access to the database is allowed. (10 marks)

166. b) By using illustrations, state the difference between hierarchical and network database

167. models (10 marks) -- 3 of 3 --

168. Distinguish between a

169. c) Explain the following database models:

170. i) Relational Model (3 marks)

171. ii) Object oriented model (3 marks)

172. d) Explain the following phases of database development

173. i) Evaluation and Selection ii) Requirements Analysis

174. iii) Implementation (6 marks)

175. f) Draw an entity relationship diagram for Machakos level five hospital that captures any

176. four entities. (4 marks)

177. a) The table below represents details of an entity to be modelled in a database. Use it to answer the questions that follow. Order Date Customer Items

178. 22/8/2021 Mark Okode Computer, Printer, Pen -- 1 of 2 -- Examination Irregularity is punishable by expulsion Page 2 of 2 Normalize this table into:

179. Explain and

180. iii) Third Normal Form (3 marks)

181. b) Explain three differences between Oracle DBMS and SQL database. (6 marks)

182. c) Distinguish between data definition language and data manipulation language as used in

183. SQL. (4 marks)

184. a) Database design is an important stage in development:

185. i) Discuss the process of designing a typical database. (6 marks)

186. b) James has been asked to create a relationship between two tables in MS-Access. Explain

187. the process he is likely to follow to accomplish this task. (4 marks)

188. Analyze

189. a) Mark has been assigned the task of designing tables to be used in a given database. Explain three reasons why he would choose to use entity relationship model as opposed

190. to other models. (6 marks)

191. Explain three

192. schema that can be used for this purpose. (6 marks)

193. a) Explain the function of each of the clauses in the SELECT statement in SQL

194. i) From ii) Where

195. iii) Group By (6 marks)

196. i) Create a database called MksU (2 marks)

197. Create

198. Identify the appropriate data

199. Select all Female Lect

200. a) List four competencies which a database administrator should have to do his work effectively. (4

201. b) Given user Brian and user Ben are transacting and updating record in the same table, with the aid of a diagram, illustrate a lost update as a result of an inconsistent read. (12

202. c) As an organization grows, there is need to establish a data warehouse. State four key roles

203. of a data ware house administrator. (4 marks)

204. use to seal the loopholes. (10 marks)

205. Please provide the

206. a) Discuss four database failure responses under the following circumstances. (8 marks) i. Aborted transactions ii. Incorrect data -- 1 of 2 -- Examination Irregularity is punishable by expulsion Page 2 of 2 iii. System failure (database intact) iv. Database destruction

207. management (4 marks)

208. d) Differentiate shared lock and an exclusive lock. (4 marks)

209. QUESTION FIVE

210. a) Explain five advantages of database management system over file management system.

211. c) Discuss four components of a database environment (4 marks)

212. d) Using clear examples, illustrate the application of three sql languages (6 marks)

213. e) Illustrate two approaches of updating data in data to a table in Ms. Access (4 marks)

214. concurrent execution of several transactions and database consistency (6 marks) -- 1 of 3 -- Examination Irregularity is punishable by expulsion Page 2 of 3

215. b) Study the following table and answer the questions that follow.

216. Normalize the above table

217. State the

218. c) Out of the 3NF table(s), generate a SQL script that displays student number, name, campus,

219. department, and class using an equijoin. (7 marks)

220. a) Justify the importance of adding indexes in a table (2 marks)

221. List the

222. c) Study the following table and answer the following questions: P_ID F_Name S_Name P_Category P_DOB Tel _Number M_Status HomeAddress

223. Write the SQL script to do the following:

224. Here's the

225. 233-2250 Narok) while applying lock to the entire table (3 marks)

226. iii) Displays all patients below 18 years (3 marks)

227. iv) Adds a constraint to avoid repetition of PatientID (4 marks)

228. List MS

229. b) Justify the importance of a database administrator in an ERP system (4 marks)

230. c) Highlight four actions involved in a checkpoint in a database system. (4 marks)

231. d) Illustrate four methods applied in ensuring security of a database system (8 marks) Std_no Name, Tel_No Campus, Address , Town Department HoD Office_Tel Program Class

232. 401 Jane, 0721545 Main, 45645 Kisii CIT Mr. Mwai 53337 Computer science J17-2022 402 Kamau, 077120 Main, 34242 Keroka ENG Mr. Tom 53341 Civil Eng. J12-2022

233. 403 Kioko,0721125 Main, 56756 Nairobi CIT Mr. Mwai 53337 Computer science J17-2022 404 Lenku, 072564 Town, 3456 Kericho ENG Mr. Tom 53352 Civil Eng. J12-2022 521 Linet, 0221525 Town, 1112 Makueni HOSP Mr. Lewa 53389 Tourism T12-2022 -- 2 of 3 -- Examination Irregularity is punishable by expulsion Page 3 of 3

234. a) Define concurrency in a database transaction (2 marks)

235. b) Simulate the following occurrences in a database system.

236. i) Lost Update (4 marks)

237. ii) Uncommitted dependency (4 marks)

238. iii) Inconsistent analysis (4 marks)

239. a) List five database management system software used in industry today (5 marks)

240. c) Discuss five reasons why you would advise a company to go for a database management

241. system other any other data management system. (5 marks)

242. e) Illustrate the step by step process of generating a report from Ms access table. (4 marks)

243. c) Discuss the three levels of schema architecture. (6 marks)

244. d) Differentiate the following (4 marks) i. Logical and physical data independence ii. Single user and multi user systems.

245. c) Study the following table and answer the following questions: Index PatientID PatientName PatientCategory DOB MaritalStatus DisabiltyType HomeCounty

246. iii) Displays all patients with disabilities in the table (3 marks)

247. a) Define a database model (2 marks)

248. These

249. b) Study the following table and answer the questions that follow. Std_no Name, Tel_No Campus, Town Department HoD Office_Tel 401 Jane, 0721545 Main, Kenya CIT Mr. X 53337 402 Kamau, 077120 Main, Kenya ENG Mr. Y 53341 403 Kioko,0721125 Main, Nairobi CIT Mr. X 53337 404 Lenku, 072564 Town, Kericho ENG Mr. Y 53337

250. c) Give the difference between delete anomaly and update anomaly. (4 marks)

251. d) Explain the following database concepts i. OLTP

252. ii. OLAP (4 marks)

253. e) Briefly describe three possible attacks to a database system (6 marks)

254. f) Draw an Entity Relationship Diagram for MU that captures the following entities: Students.

255. Lecturers, Library and Computer Lab. Include 2 attributes for each entity. (4 marks)

256. b) Explain four file organization methods that can be used in databases. (8 marks)

257. a) Distinguish between the following: i. Primary key and foreign key

258. ii. Inner join and right join (8 marks)

259. c) Discuss three constraints that can be used in database tables (6 marks) -- 2 of 2 --

260. i. Atomicity (2 marks)

261. ii. Aggregation (2 marks)

262. iii. ERD (2 marks)

263. Define 'Data

264. e) Describe three objectives of the selection of a data type for databases. (6 marks)

265. f) Write an SQL SELECT statement to display all the columns of the STUDENT table but

266. only those rows where the Grade column is greater than or equal to 90. (3 marks)

267. g) Explain what needs to happen to convert a relation to third normal form (6 marks)

268. a) Write an SQL SELECT statement to count the number of rows in STUDENT table and

269. display the result with the label NumStudents. (3 marks)

270. b) Discuss the alternative terminology that is used in the relational model. (4 marks)

271. Define domain constraints and

272. a) Explain why normalized tables require more complex SQL when SQL statements are

273. used in application programs (4 marks)

274. (4 marks) Study the following database table that stores student’s records and answer the questions that follow:- REG_NO F_NAME L_NAME COURSE GENDER D_O_B 1/2016 John Mutuku DCS Male 13/6/1990 2/2016 Steve kipkorir DCS Male 13/3/1985 3/2016 Susan Mutua CIT Female 19/11/1986 4/2016 Steve Kin’gori DBIT Male 1/3/1978

275. i) Display all students where the results will only contain the columns RGE_NO,

276. L_NAME and F_NAME. (2 marks)

277. ii) To produce the L+NAME and F_NAME combined as Student name (3 marks)

278. iii) To find the details of All male students in the course DCS. (2 marks)

279. iv) To produce the total number of students per course (2 marks)

280. v) Get a list of all the male students born between the year 1980 and 1989. (3 marks)

281. b) Describe the difference between homogeneous and heterogeneous distributed database

282. c) Explain TWO Applications of Distributed Databases (2 marks) -- 2 of 3 -- Examination Irregularity is punishable by expulsion Page 3 of 3

283. d) You have two tables, EMPLOYEE and COMPUTER that are in a one-to-one

284. relationship. The foreign key is EmpNumber in COMPUTER which references EmpNumber as the primary key of EMPLOYEE. Explain what must be done to convert

285. the one-to-one EMPLOYEE-COMPUTER relationship to a one-to-many relationship

286. where one employee can have more than one computer. (6 marks)

287. b) The Database Life Cycle (DBLC) contains six phases, Analyze what happens in phase

288. one. (6 marks)

289. c) Discuss FOUR disadvantages of a standard language such as SQL (8 marks) -- 3 of 3 --